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Application of the MOF [Eu(BTC)] as a Luminescent Marker for GSR: A Synthesis, Characterization and Toxicity Study.

Marcella A. M. Lucena,^{a,b} Marina F. L. Oliveira,^d Aline M. Arouca,^b Márcio Talhavini,^f Eduardo A. Ferreira,^e Severino Alves Júnior,^c Fabiane H. Veiga-Souza,^{d,e} Ingrid T. Weber^{a,b,}*

^aPGMTR – CCEN, Universidade Federal de Pernambuco–UFPE, 50740-540, Recife – PE, Brazil

^bLIMA, Instituto de Química, Universidade de Brasília-UnB, 70904-970, Brasília – DF, Brazil

^cBSTR, Departamento de Química Fundamental-DQF, Universidade Federal de Pernambuco–UFPE, 50740-540 Recife – PE, Brazil

^dLBQP, Instituto de Biologia, Universidade de Brasília-UnB, 70910-900, Brasília, DF – Brazil

^eFaculdade de Ceilândia, Universidade de Brasília-UnB, 72220-275, Brasília, DF – Brazil

^fInstituto Nacional de Criminalística, Polícia Federal, SAIS Quadra 07 Lote 23, 70610-200 Brasília, DF, Brazil.

KEYWORDS. luminescent markers, gunshot residues, MOFs, toxicity, OECD 423.

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2
3 ABSTRACT. The 3D metal-organic framework (MOF) [Eu(BTC)] (where BTC = trimesic acid)
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5 was synthesized in 20 minutes by a microwave-assisted hydrothermal method with a yield of
6
7 89%. A structural and spectroscopic study, performed by X-ray diffraction, thermogravimetry
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9 and photoluminescence spectroscopy, showed that this framework has high crystallinity, thermal
10
11 stability and luminescence. This MOF had a red-orange luminescence when excited with UV
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13 radiation ($\lambda = 254$ nm) and a high potential for use as a luminescent marker for gunshot residues
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15 (GSR). When added to 9-mm non-toxic ammunition (NTA), it greatly improved quality of the
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17 crime scene investigation, allowing for direct visualization of the luminescent GSR on the
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19 shooter's hand and firearm and at the firing range using only a portable UV lamp. The marked
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21 luminescent GSR (LGSR) was easily collected and characterized by non-destructive techniques,
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23 including VSC and SEM/EDS, wherein the presence of Eu^{3+} ions was confirmed. Furthermore,
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25 the oral acute toxicity of this MOF was assessed in adult female Wistar rats using the OECD 423
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27 guidelines. This study classified the MOF [Eu(BTC)] in a less toxic GHS category (category 5),
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29 with an LD_{50} of 5000 mg/kg, ensuring a wide security range for its application.
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41 INTRODUCTION

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43 Gunshot residue (GSR) characterization is a point of primary interest in forensic science because
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45 it can provide essential information about crimes, which helps, for example, in identifying
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47 suspects and understanding the crime dynamics.^{1,2} Daily, experts need to investigate crimes that
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49 involve firearms, but the identification of GSR is not a trivial task. Traditionally, the American
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51 Society for Testing and Materials (ASTM) defines GSR from conventional ammunition as
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53 spheroidal particles containing Sb, Ba and Pb.³ Although different techniques can detect these
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metals in nanograms (as ICP-OES, AAS and ICP-MS),⁴⁻¹⁰ the ASTM recommends the use of scanning electron microscopy coupled with energy dispersive X-Ray (SEM/EDS), which can provide morphological and compositional information that ensures these metallic elements are present in the same spheroid particle.³ However, this scenario with well-established protocols changed when a study showed that frequent shooters had high levels of lead in their blood¹¹ and lead-free ammunitions (or non-toxic ammunitions - NTA) were developed. Despite the advantages of this new type of ammunition, the lack of characteristic elements made the traditional detection methodologies ineffective.^{12,13}

In this context, different proposals have emerged for GSR identification, mainly with respect to the analysis of the organic part of GSR (OGSR).¹⁴⁻¹⁷ In López-López et al.,¹⁷ for example, OGSR particles of six lead-free ammunitions were differentiated as function of the type of stabilizer in the gunpowder formulation (diphenylamine or centralite) using Raman spectroscopy. Meanwhile, some studies suggested the addition of chemical markers to the primer or gunpowder of lead-free ammunition to identify GSR and codify ammunition.^{12,18} Our group dedicated efforts to develop suitable luminescent markers for GSR identification.¹⁹

A great advantage of the luminescent markers is they make the GSR particles visible, allowing for direct visualization *in loco* using only an UV lamp.²⁰⁻²⁴ This feature facilitates crime scene investigation, the collection of luminescent GSR (LGSR) and lab analysis. Additionally, the lanthanide ions present in luminescent markers acts as chemical markers, allowing for differentiation of the LGSR from other environmental or occupational residues. Finally, non-destructive methods can be used to characterize the LGSR.²⁰⁻²⁵

The first materials with promising results as luminescent markers for GSR were the lanthanide metal-organic frameworks (Ln-MOFs) [Ln(DPA)(HDP)] (DPA = dipicolinic acid and Ln =

Eu³⁺, Tb³⁺, Dy³⁺ and Yb³⁺/Tb³⁺)^{20,21,23,24} and the oxides ZnAl₂O₄:Ln (Ln = Eu³⁺ and Tb³⁺).²²

These markers allowed for direct visualization of the LGSR on surfaces close to the shooting location, such as the shooter's hand, firearm, firing range, cartridges, target, etc.

Recently, the efficiency of these markers in circumstances that mimic real situations was reported; these conditions include shots effectuated inside/against vehicles, simulated victims (pork leg), the transference of LGSR to objects handled by the shooter, collection in unconventional locations, and the persistence of the LGSR to handwashing.²³ In all situations, the luminescent markers provided essential information that can help forensic experts investigate a crime scene.

Despite advances, the application of this new tool is subject to knowledge of the toxicity of these markers, which will be in direct contact with shooters. Although a wide range of MOFs have biomedical applications,^{26–34} little is known about these materials. Furthermore, the development of new markers can help especially in the process of ammunition traceability. In this work, the following are presented: the fast synthesis of the luminescent Ln-MOF [Eu(BTC)] (where BTC = trimesic acid), isostructural to the MIL-78,³⁵ its efficiency as luminescent marker for GSR and its acute oral toxicity in rats. To the best of our knowledge, applications for this MOF have not yet been reported and nothing is known about its toxicity.

EXPERIMENTAL SECTION

Synthesis and characterization of the MOF [Eu(BTC)]: With MW-assisted hydrothermal synthesis, 0.175 mmol of Eu₂O₃ (Sigma Aldrich, 99.5%), 0.35 mmol of BTC (Sigma Aldrich, 95%) and 5 mL of deionized water were added in a 10-mL quartz reactor. All chemicals were used as purchased without purification. The reaction was performed in a CEM Corporation

microwave (Discover Proteomics®) with a maximum power of 100 W, for 20 minutes, at 150°C and autogenous pressure. The powder obtained was washed three times with deionized water and then with ketone, dried at 75°C, and characterized by X-ray diffraction (XRD, Bruker D8 Advanced), photoluminescence spectroscopy (PL, Spectrofluorometer Fluorolog Horiba Jobin Yvon FL3-22) and thermogravimetric analysis (TGA/DTG-60, Shimadzu).

Test of a luminescent marker for GSR: To perform the tests of a luminescent marker for GSR, the marker was add to the gunpowder (10 wt%) of 9-mm lead-free ammunitions according to the methodology described in previous studies.^{20,22} Then, 4 shots were performed using a GLOCK 17 pistol at an indoor shooting range in the ballistic service of NIC/PDF (National Institute of Criminalistics of Brazilian Federal Police). The LGSR deposited on the shooter's hands and firearm was visualized under UV radiation ($\lambda = 254$ nm) and collected with stubs covered with double-sided adhesive conductive carbon tape. The collected LGSR was analyzed with a Video Spectral Comparator (VSC, Foster & Freeman VSC® 6000/HS) and SEM/EDS (FEI Quanta 200 FEG).

Toxicity analysis: Wistar rats (females, 170-190 g in body weight, 8 weeks old, São Paulo University, São Paulo, Brazil) were housed, three per cage, with standard rat chow and water *ad libitum*. Housing was in a temperature-controlled room ($22 \pm 3^\circ\text{C}$) with a 12 h light-dark cycle (lights on at 07:00 AM). All rats were allowed 1 week of acclimation to the colony rooms before starting the procedures. Care and use of the animals were in full compliance with the Ethical Principles in Animal Research adopted by the National Council for the Control of Animal Experimentation (CONCEA), and this study was previously approved by the Animal Research Ethics Committee of the University of Brasilia (Protocol nr. 131598/2015).

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3 The acute oral toxicity of the MOF [Eu(BTC)] was evaluated in rats, with a single-dose
4 exposure, according to the Guideline 423 of the Organisation for Economic Cooperation and
5 Development (OECD).³⁶ These guidelines employ a stepwise procedure with 3 animals of a
6 single sex per step, wherein the next dose is set based on the mortality and/or moribund states
7 presented by the animals on the previous dose.
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11 To monitor the body temperature, battery-operated transmitters (Subcue dataloggers, Calgary,
12 Canada) were implanted in the abdominal cavity of the rats under anesthesia with a mixture of
13 ketamine and xylazine (60 and 10 mg/kg, respectively, i.p.). A 1-week recovery period was
14 allowed before experimentation.
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17 For oral administration, animals were fasted overnight prior to dosing (just food) and 4 hours
18 post-dosing (food and water). The MOF was suspended in saline solution (30 or 100 mg/mL) and
19 administered in a single dose by gavage through a polyethylene cannula (PE-50, 4 cm length)
20 connected to a needle placed in a 5-mL plastic syringe. The administered volume ranged from
21 1.7 to 4.0 mL according to the animal body weight and the dose. The starting dose was 300
22 mg/kg, as recommend by guidelines when there is no information about the substance toxicity.
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24 The control group received saline solution.
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27 After administration, rats were continuously monitored for 1 h after dosing. They were
28 periodically observed during the first 24 hours (with special attention to the first 6 hours) and
29 daily for the next 14 days. All observations were recorded, including the body weight and water
30 and food intake. At the end of the experiment, animals were anesthetized (ketamine 120 mg/kg
31 and xylazine 20 mg/kg) to collect blood from the inferior vena cava (using heparin as
32 anticoagulant) and euthanized by decapitation. All animals were subjected to gross necropsy.
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34 The weights of the heart, spleen, kidney and liver of each rat were recorded. According to the
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guidelines, histopathologic analysis is only recommend in organs showing evidence of gross pathology.³⁶

The collected blood was centrifuged at 5000 g for 5 minutes and the serum was collected and stored at -18°C until analysis. Biochemical parameters related to the hepatic (alanine aminotransferase – ALT) and renal function (urea) were performed with spectrophotometric methods (LAMBDA™ 35 Series UV/Vis Spectrophotometers, PerkinElmer®) using a commercial kit (LABTEST, Diagnostica S.A., Minas Gerais, Brazil).

The statistical analysis of the biochemical parameters was performed using GraphPad Prism software. Comparisons among different groups was performed using the analysis of variance test (ANOVA). All data are expressed as the mean ± standard error of mean (SEM); p values less than 0.05 were considered significant.

RESULTS AND DISCUSSION

Structural and Spectroscopic Study of the MOF [Eu(BTC)]: The MOF MIL-78 (or [Y_{1-x}Ln_x(BTC)]) was synthetized by Serré *et. al.*³⁵ by a conventional hydrothermal method at 220°C for 5 days. Singh *et al.*³⁷ also synthetized the framework [Y(BTC)] by a mechanochemical method in 7 hours using yttrium hydride as the starting material. However, in this work, the use of the microwave (MW)-assisted hydrothermal method to quickly synthesize the MOF [Eu(BTC)] is presented.

In the MW-assisted method, the microwave radiation interacts directly with the dipole moment of polar molecules, resulting in a fast increase in the rotational and vibrational energies of the molecules as well as in the temperature and pressure of the system.^{38–40} For this reason, when compared to the methods already used to synthesize the MIL-78, the MW-assisted method

represented a significant reduction in the time consumption. The XRD in figure 1 shows that a MOF that was isostructural to the MIL-78 was obtained in only 20 minutes with this method (up to 360 times faster than the hydrothermal conventional method). Moreover, this synthesis presented a high and reproducible yield of $89 \pm 1.5\%$ (syntheses were performed in triplicate).

The MOF [Eu(BTC)] crystallized in monoclinic symmetry, where each Eu^{3+} ion is coordinated by eight oxygen atoms from six BTC ligands, and the coordination polyhedron can be represented as a dodecahedron trigonal prism (D_{2d}) (figure 1.c). BTC ligand assumes two different coordination modes, bidentate bridging and tridentate bridging-chelating (figure 1.d),⁴¹ which enable the 3D growth of this MOF.

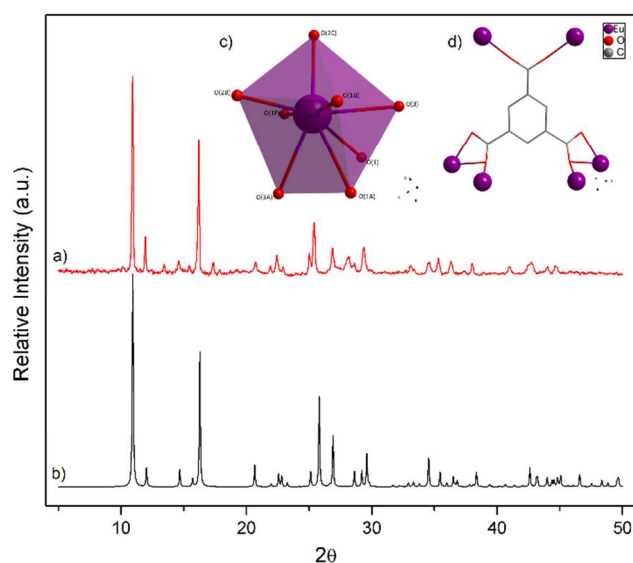


Figure 1. XRD pattern a) of the MOF [Eu(BTC)] and b) calculated for the MIL-78, c) Eu^{3+} coordination polyhedron and d) BTC coordination modes (figures c and d were created using crystallographic information (CIF) of the MOF MIL-78).

The thermogravimetric analysis (figure 2) showed that this framework presents with high thermal stability, which is a positive point for its application as a luminescent marker. According

to Rezende *et al.*,⁴² the free-BTC ligand degrades between 300°C and 450°C. However, once it is present in the MOF [Eu(BTC)], its degradation occurs at higher temperatures (between 430°C and 600°C) because lanthanide ions and ligand molecules are strongly connected through Ln-O-C bonds, forming rigid and stable structures.⁴³ The total mass loss of 51% is in good agreement with the calculated value for the degradation of organics.^{35,37}

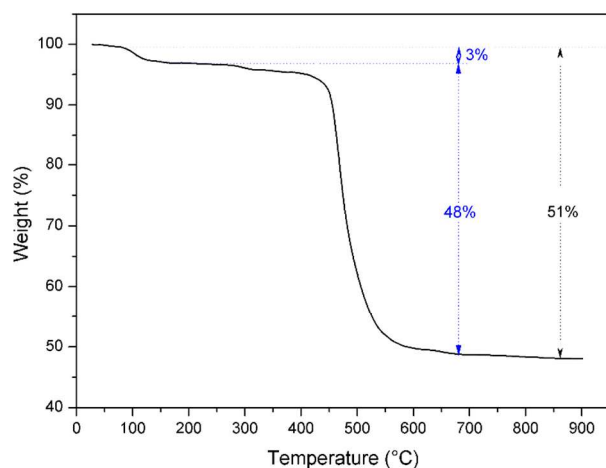


Figure 2. Thermogravimetry analysis (TGA) curves registered under dynamic N₂ atmosphere of the MOF [Eu(BTC)].

Excitation and emission spectra were obtained at room temperature (figure 3). The excitation spectrum (black line, obtained monitoring the emission at 617 nm) showed a broad band in 303 nm relative to the π - π^* transitions of the ligand⁴⁴ as well as narrow and less intense bands relative to the f-f transitions of Eu³⁺.⁴⁵ In this case, the antenna effect, which is normally more efficient than the direct excitation of the lanthanide ions, was the main excitation mechanism. The emission spectrum (red line, obtained by exciting at 303 nm) showed 5D_0 - 7F_J ($J = 0 - 4$) transitions of Eu³⁺,⁴⁶⁻⁴⁸ being the hypersensitive electric-dipole transition 5D_0 - 7F_2 the most

intense. This MOF emitting an intense orange-red color, characterized with the CIE 1931 coordinates as (0.63, 0.35) (figure S1 – supplementary information).

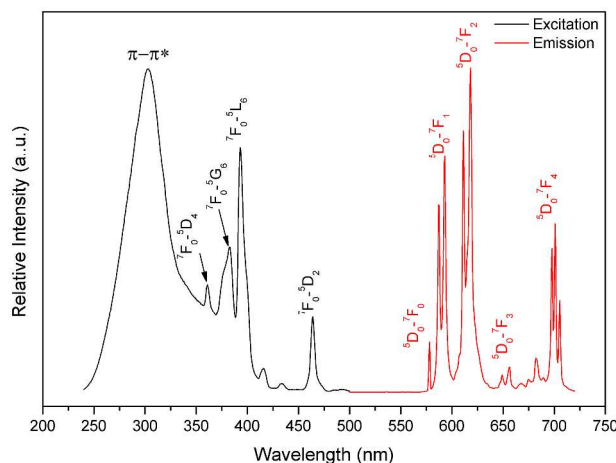


Figure 3. Excitation and emission spectra of the MOF [Eu(BTC)].

Structural and spectroscopic information can be obtained by calculating the asymmetry ratio (R) and Judd–Ofelt intensity parameters Ω_2 and Ω_4 from the emission spectrum of Eu^{3+} .⁴⁶ These parameters, as well as the radiative (A_{rad}) and non-radiative (A_{nrad}) decay rates, quantum efficiency (η) and lifetime are shown in table 1. The low values of the asymmetric ratio (R) and intensity parameters (Ω_2) revealed high symmetry and slightly polarizable chemical environments around the Eu^{3+} ions.⁴⁹ Moreover, a rigid structural arrangement was demonstrated by the low values of Ω_4 .⁵⁰ A rigid arrangement is desirable because it improves the luminescent features (by reducing non-radiative decays) and enlarges the chemical and thermal stability, which are extremely important for a MOF to be used as a GSR marker.

Furthermore, this MOF has a long lifetime (0.93 ms), which can be associated with the absence of water molecules in the Eu^{3+} coordination sphere. Coordinated water acts as quenching channels that favor non-radiative decays. Other Eu-BTC MOFs that have coordinated water have

shorter lifetimes ($[\text{Eu}(\text{BTC})(\text{H}_2\text{O})_6] = 0,23 \text{ ms}^{51}$ and $[\text{Eu}(\text{BTC})(\text{H}_2\text{O})] = 0,84 \text{ ms}^{52}$) and higher non-radiative decay rates ($[\text{Eu}(\text{BTC})(\text{H}_2\text{O})_6] = 3820 \text{ s}^{-1} \text{ }^{51}$). These features gave the MOF $[\text{Eu}(\text{BTC})]$ a higher quantum efficiency (almost 20%).

Table 1. Lifetime (τ), asymmetric ratio (R), Judd–Ofelt intensity parameters Ω_2 and Ω_4 , radiative (A_{rad}) and non-radiative (A_{nrad}) decay rates and the quantum efficiency (η) of the MOF $[\text{Eu}(\text{BTC})]$.

	τ (ms)	R	Ω_2 (10^{-20} cm^2)	Ω_4 (10^{-20} cm^2)	A_{rad} (s^{-1})	A_{nrad} (s^{-1})	η (%)
$[\text{Eu}(\text{BTC})]$	0,93	1,84	3,29	3,73	210	870	19,5

Performance of the MOF as a luminescent marker for GSR: Once characterized, the MOF $[\text{Eu}(\text{BTC})]$ was introduced in gunpowder of 9-mm NTA, and shots were fired using a GLOCK 17 pistol. Figures 4.a and 4.b show the LGSR on the shooter hand’s and firearm under UV radiation ($\lambda = 254 \text{ nm}$). Despite the micrometric size of these particles, the LGSR was visualized using only a portable UV lamp. This procedure simplifies the analysis of the suspect and firearm as well as enable directed and efficient sample collection, improving the possibility of finding a GSR particle and reducing the false-negative probability, as shown in figure 4.c-d. Moreover, as luminescent particles are not regularly found on the hands or in the environment, these particles are strong indicative of LGSR.

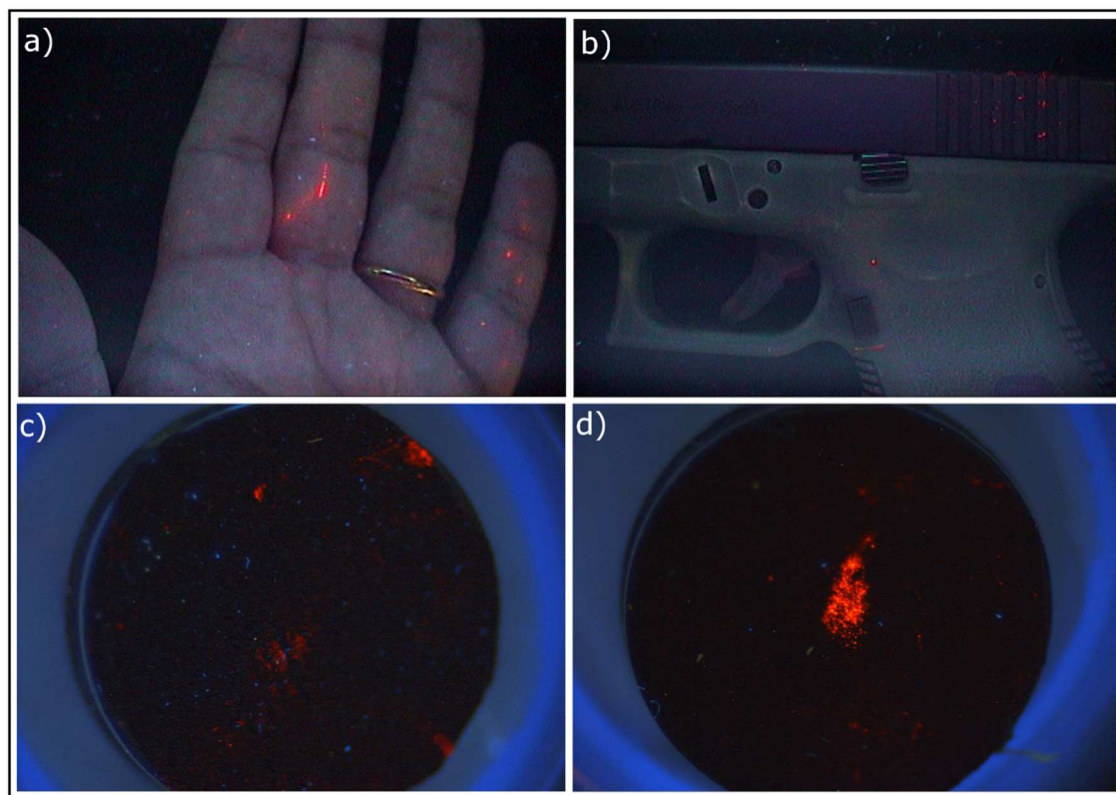


Figure 4. VSC images ($\lambda = 254$ nm) showing LGSR particles on a) the shooter's hand and b) the firearm as well as collected on c) the shooter's hand and d) the firearm.

After collection, the LGSR particles were analyzed by VSC ($\lambda = 254$ nm), to acquire emission spectra, and by SEM/EDS (figure 5). The emission spectra showed main transitions of Eu^{3+} , ensuring that the observed red color is a product of the Eu^{3+} electronic transitions instead of being from other nonspecific organic or environmental particles. Compared to the pure marker, a reduction in the site symmetry was observed (the asymmetric ratio R), which could be from the high temperature or pressure of the shots. However, this did not affect the marker visualization. In SEM/EDS analysis, particles with morphology that was similar to the pure marker were found, facilitating the analysis. The presence of europium in these particles was confirmed by EDS. Recently, the use of a lanthanide ion as a chemical marker of NTA used for some

European police forces was reported.⁵³ In this case, gadolinium was used as a chemical marker.^{53,54} However, this ion does not enjoy luminescent features, which is unlike Eu^{3+} . As a result, Eu^{3+} has a great advantage of acting simultaneously as a chemical marker, given its unusual occurrence, and an optical marker due to its strong luminescence in the Eu-MOF. Furthermore, in both cases (Gd or Eu-MOF), the methodology that was already adopted to characterize the conventional GSR (SEM/EDS) remains efficient with these markers, allowing police forces to continue using the usual procedures.

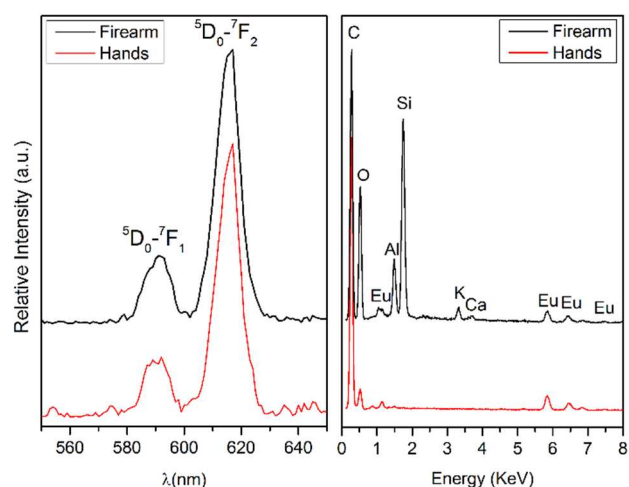


Figure 5. Emission spectra (left) and EDS (right) of the LGSR collected on the firearm (black) and shooter's hands (red) after shots with marked ammunition.

Toxicity study of the MOF [Eu(BTC)]: The acute oral toxicity of the MOF [Eu(BTC)] was assessed in young adult female Wistar rats with a single-dose exposure according to the OECD Guideline 423.³⁶ Following the protocol steps, four groups of three rats were evaluated, named G1, G2, G3 and G4, which were treated with 300, 300 (repetition), 2000 and 2000 (repetition) mg/kg, respectively, as shown in figure 6. There was also a control group (CG). At all doses,

none of the animals demonstrated clinical toxicity, abnormal behavior or died. Additionally, there were no registered changes in the body and organ weights (heart, liver, spleen and kidneys) or water and food consumption compared to the control group (figure 7). Thus, the results indicated that the oral LD₅₀ of the MOF [Eu(BTC)] was 5000 mg/kg body weight, which was classified as category 5 of the Globally Harmonized System (GHS) and suggested the compound had good safety with minimal toxicity.

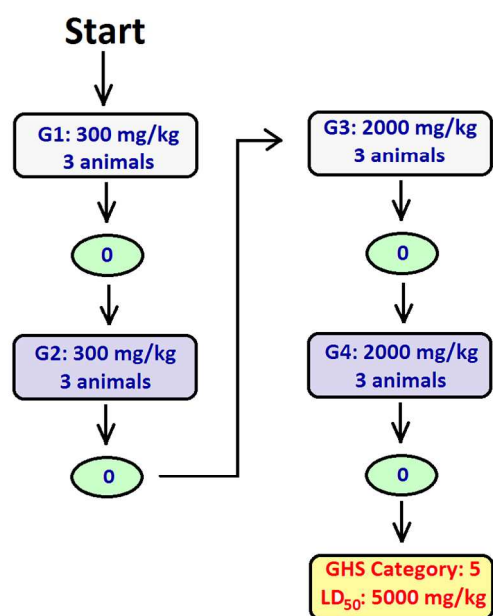


Figure 6. Steps followed by OECD guideline 423 to determine the LD₅₀ and GSH category. The circles indicate the number of dead or moribund animals by step.

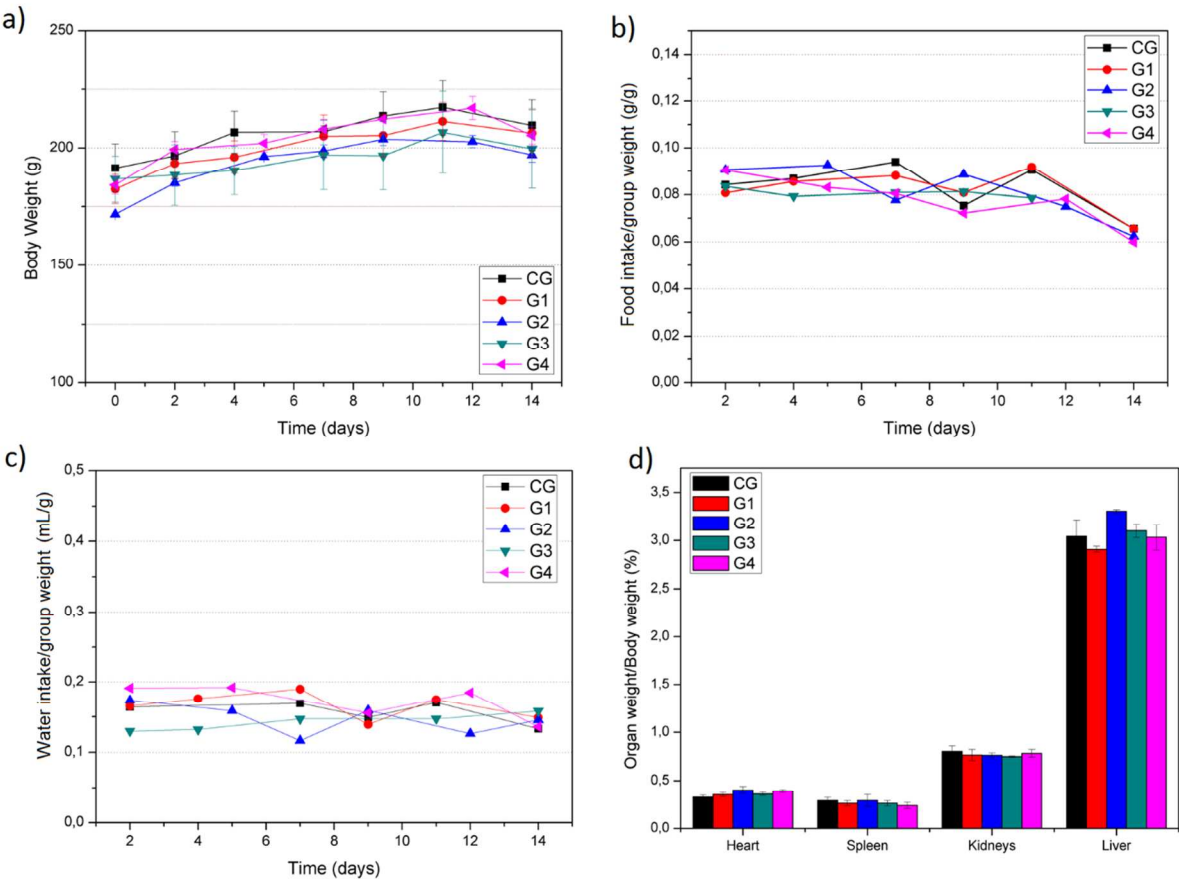


Figure 7. a) Evolution of the rat weight (mean $\pm$ SD), b) mean food and (c) water intake per group; d) organ weight as a function of the body weight in each group (mean $\pm$ SD).

In addition, the core body temperature of the rats was monitored by implanted transmitters 4 hours before dosing until the end of experiments. It is known that deviations in the body temperature can serve as indicators of the clinical conditions of an animal. For example, in severe hypothermia (when the body temperature drops below 27°C), the associated neuromuscular, cardiovascular, hematological, and respiratory changes can be fatal.⁵⁵ Beyond a body temperature of 42°C, cytotoxicity occurs with protein denaturation, resulting in end-organ failure and neuronal impairment.⁵⁶

The temperature variation pattern over days can be described as a sine-like function wherein maximum temperature values occur during the night, which is due the nocturnal habits of the rodents (figure 8). In this case, some peaks out of the curve were registered in all groups (including control group) and were attributed to stress-induced hyperthermia (SIH)⁵⁷ caused by handling the animals. Thus, the absence of a pattern of changes in the body temperature compared to the control group highlights the absence of deteriorating clinical conditions in the organisms due to the marker administration. Furthermore, there were no pathological changes in the organs at necropsy.

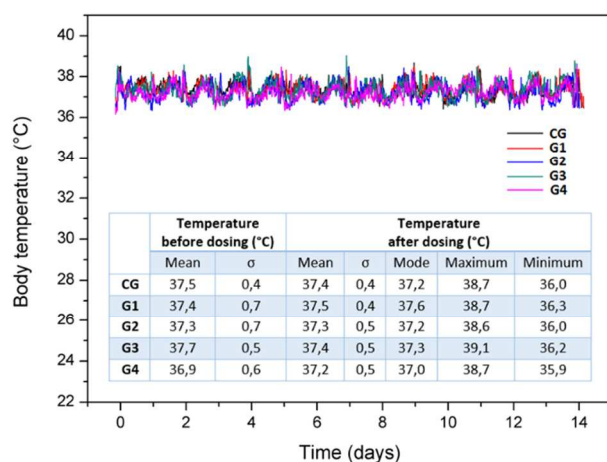


Figure 8. Time course of core body temperature for the G1, G2, G3, G4 and CG groups. The table shows the mean temperature of the groups 4 hours before dosing and 14 days after dosing; it also shows the maximum, minimum and the mode value temperatures after dosing.

Alanine aminotransferase (ALT) and urea are good indicators of the liver and kidney functions, respectively.^{58,59} The results of the biochemical analysis of the animal serum are shown in figure 9. The statistical analysis showed no difference ($p > 0.05$) for the urea and ALT concentrations between the control and treated groups. These results showed that, even for the highest dose of

2000 mg/kg, no damage to the liver or kidneys was observed 14 days after treatment, reinforcing the absence of toxicity and the GHS classification as category 5.

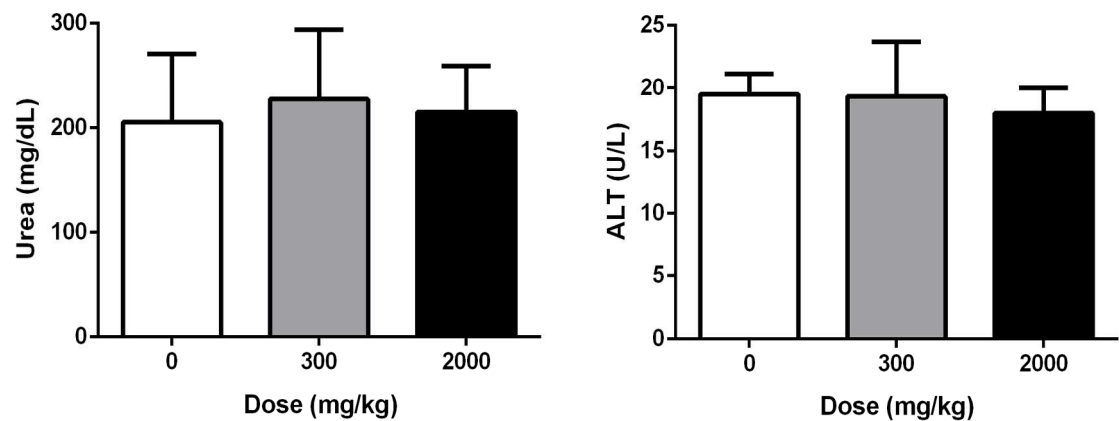


Figure 9. Biochemistry analysis of rats 14 days after treatment with the MOF [Eu(BTC)]. Values are expressed as the mean $\pm$ SEM. Dose = 0 mg/kg refers to the control group.

Comparing the toxicity of the marker [Eu(BTC)] with that of lead (Pb), which is an intrinsic chemical marker of conventional ammunitions, this MOF appears to be an efficient and non-toxic alternative to identify gunshot residues. For example, according to the National Institute for Occupational Safety and Health (NIOSH), the lowest lethal dose (LD_{LO}) of $Pb(CN)_2$ in rats (intraperitoneal) is 100 mg/kg, and the human oral lethal dose is 714 mg/kg of lead acetate (approximately 450 mg/kg of lead).⁶⁰ According to the International Program of Chemical Safety (IPCS),⁶¹ the oral LD_{50} in rats for tetraethyl and tetramethyl lead are 1.2 and 105 mg/kg, respectively, showing the high toxicity of lead compounds and the high security of the MOF [Eu(BTC)] for this application.

Other luminescent marker that showed promising results for GSR detection was the complex $Eu(PIC)_3(NMK)_3$ (PIC = picric acid and NMK = n-methyl-caprolactam) developed by Destefani

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3 *et. al.*⁶² This complex was considered as having medium toxicity ($LD_{50} = 1000$ mg/kg and GHS
4 category = 4). Comparing both markers, the MOF [Eu(BTC)] has a LD_{50} five times higher,
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6 reinforce its high security and potential application as luminescent marker for GSR.
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10 The 5000 mg/kg dose for a person of 70 kg, for example, is equivalent to 350 grams of marker,
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12 which is enough to mark more than 10 thousand ammunition rounds. This same dose is
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14 equivalent to a person of 70 kg being exposed to 233 g of marker/ m^3 for 30 min (assuming a
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16 breathing rate of 50 liters per minute and 100% of absorption).⁶⁰ To produce this atmosphere in a
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18 shooting range of 200 m^3 , for example, it would be necessary to use 46.6 kg of marker (or 1.5
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20 million shots with 10wt% of marker in 30 min), which is not achieved, thus far, in real-life
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22 situations. Finally, considering analytical reagents, 1 gram of the MOF [Eu(BTC)] costs
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24 approximately 7.65 dollars, which represents a cost of 24 cents per dollar on ammunition (using
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26 10wt% of marker).
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34 CONCLUSION

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36 The MOF [Eu(BTC)], isostructural to the MIL-78, was efficient and reproductively synthesized
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38 using a MW-assisted hydrothermal method, which represented a great reduction in the reaction
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40 time. Moreover, this MOF presented high crystallinity, thermal stability and luminescence.
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42 When incorporated into 9-mm NTA, it acted as an efficient luminescent marker, allowing for
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44 visualization of the LGSR *in loco* using only an UV lamp. These residues were easily visualized
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46 and collected in the shooters' hands and firearm. The collected particles were characterized by
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48 VSC and SEM/EDS. The emission spectra of the LGSR confirmed the luminescence results from
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50 Eu^{3+} ions and the already recommended technique to characterize conventional GSR based on
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52 SEM/EDS showed be very adequate for characterizing the LGSR. Furthermore, this framework
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was classified in the least toxic GHS category with a LD₅₀ of 5000 mg/kg, which ensures a wide security range for its application. In this way, the MOF [Eu(BTC)] was demonstrated as a non-toxic and efficient alternative for detecting and characterizing GSR from both conventional and non-toxic ammunition, facilitating crime scene investigation.

ASSOCIATED CONTENT

The following files are available free of charge.

Decay curve from the excited state ⁵D₀ to the state ⁷F₂ in the MOF [Eu(BTC)] and CIE chromaticity diagram of the MOF [Eu(BTC)] (PDF).

AUTHOR INFORMATION

Corresponding Author

itweber@unb.br

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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ABBREVIATIONS

GSR, gunshot residues; OGSR, organic gunshot residues; LGSR, luminescent gunshot residues; ASTM, American Society for Testing and Materials; SEM/EDS, Scanning Electron Microscopy coupled with Energy Dispersive X-Ray; NTA, Non-Toxic Ammunitions; ICP-OES, Inductively Coupled Plasma - Optical Emission Spectroscopy; AAS, Atomic Absorption Spectroscopy; ICP-MS, Inductively Coupled Plasma Mass Spectrometry; MOF, Metal-Organic Framework; DPA, dipicolinic acid; BTC, trimesic acid; MIL- Material of Institute Lavoisier; MW, microwave; XRD, X-Ray Diffraction; PL, Photoluminescence Spectroscopy, TGA, Thermogravimetric Analysis; VSC, Video Spectral Comparator; FEG, field emission gun; CONCEA, National Council for the Control of Animal Experimentation; OECD, Organisation for Economic Cooperation and Development; ALT, alanine aminotransferase; GHS, Globally Harmonized System (GHS); NIOSH, National Institute for Occupational Safety and Health; LD, lethal dose; IPCS, International Program of Chemical Safety.

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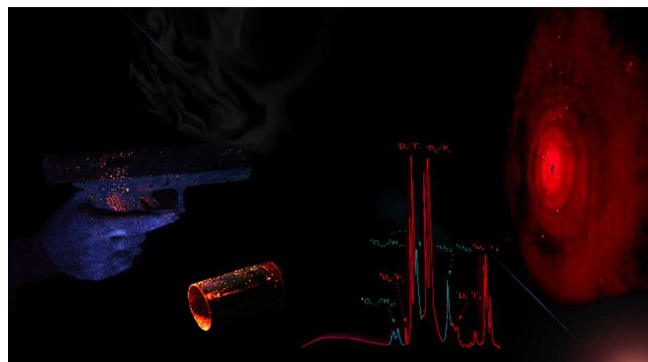
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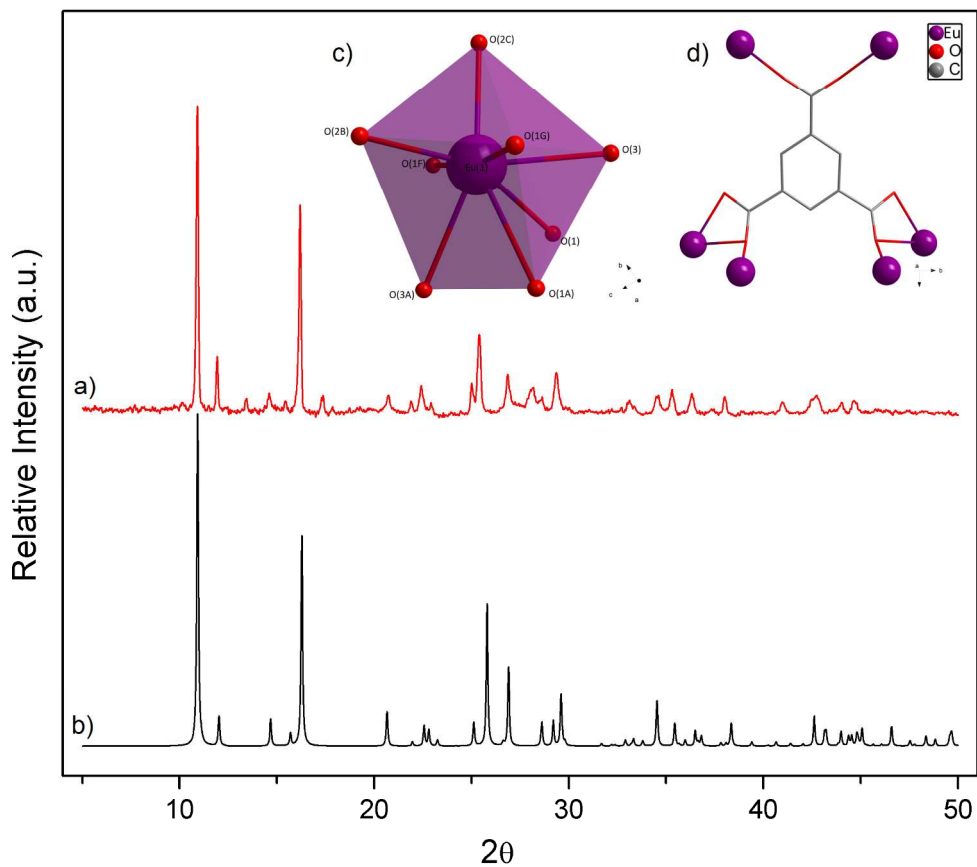


Figure 1. XRD pattern a) of the MOF [Eu(BTC)] and b) calculated for the MIL-78, c) Eu³⁺ coordination polyhedron and d) BTC coordination modes (figures c and d were created using crystallographic information (CIF) of the MOF MIL-78²⁸).

Figure 1
673x603mm (96 x 96 DPI)

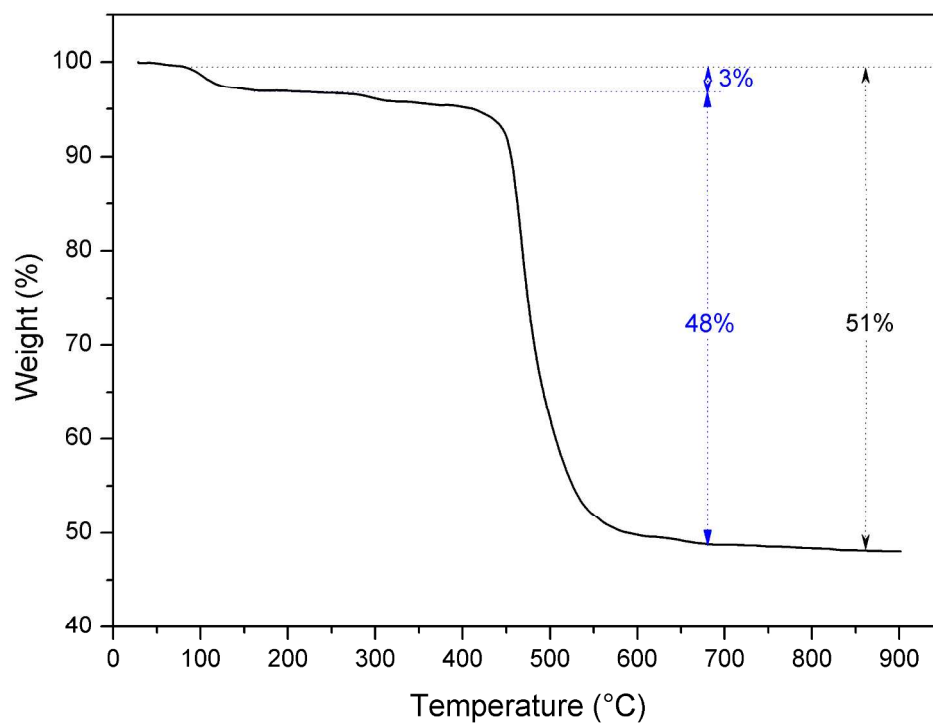


Figure 2. Thermogravimetry analysis (TGA) curves registered under dynamic N₂ atmosphere of the MOF [Eu(BTC)].

Figure 2

725x543mm (96 x 96 DPI)

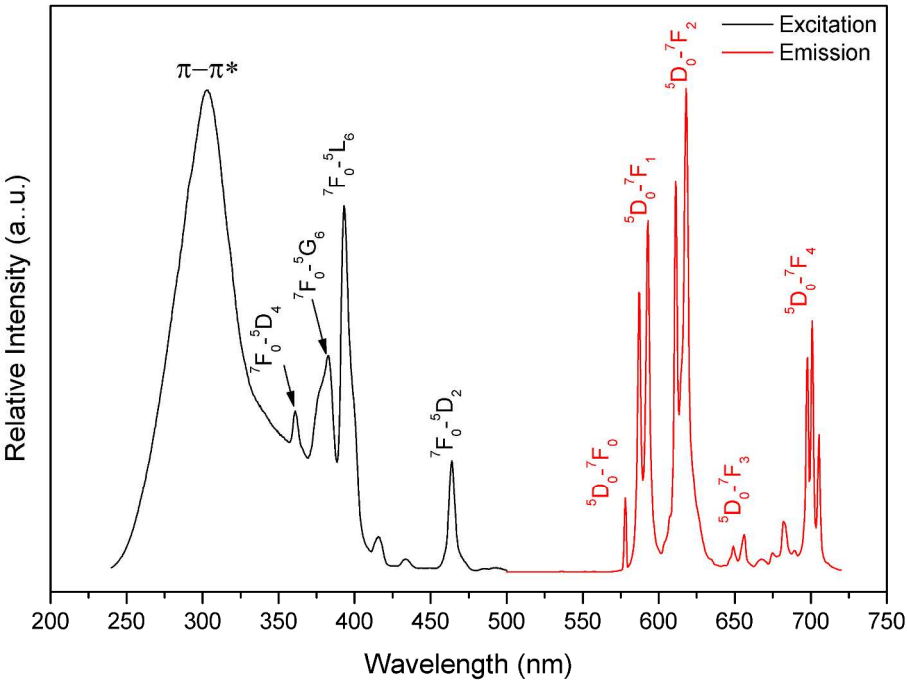


Figure 3. Excitation and emission spectra of the MOF [Eu(BTC)].
Figure 3
281x197mm (300 x 300 DPI)

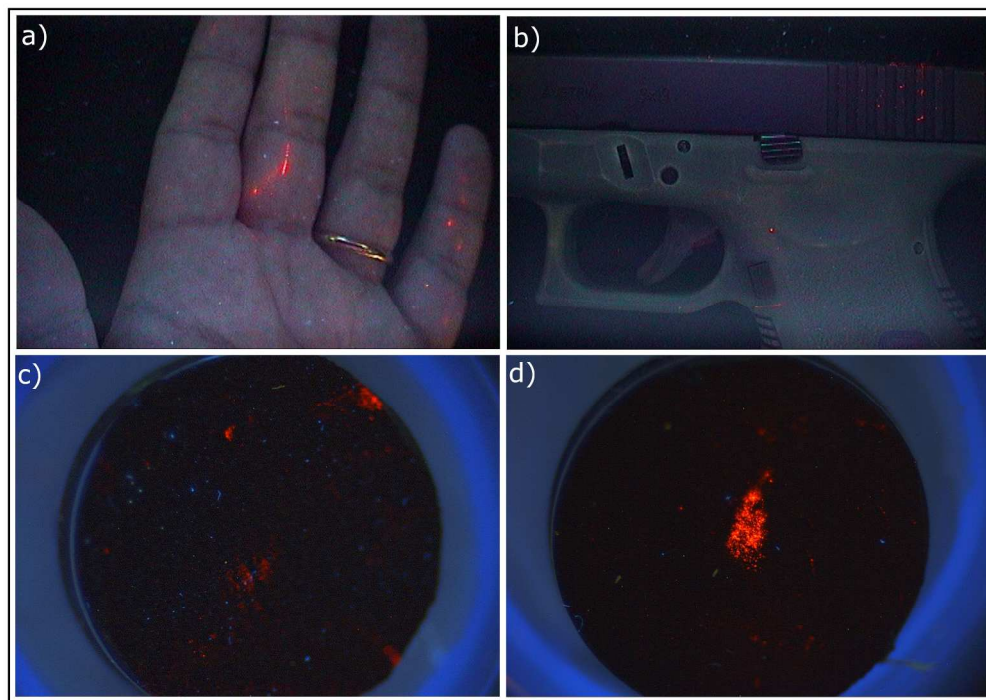


Figure 4. VSC images ($\lambda = 254$ nm) showing LGSR particles on a) the shooter's hand and b) the firearm as well as collected on c) the shooter's hand and d) the firearm.

Figure 4

706x498mm (96 x 96 DPI)

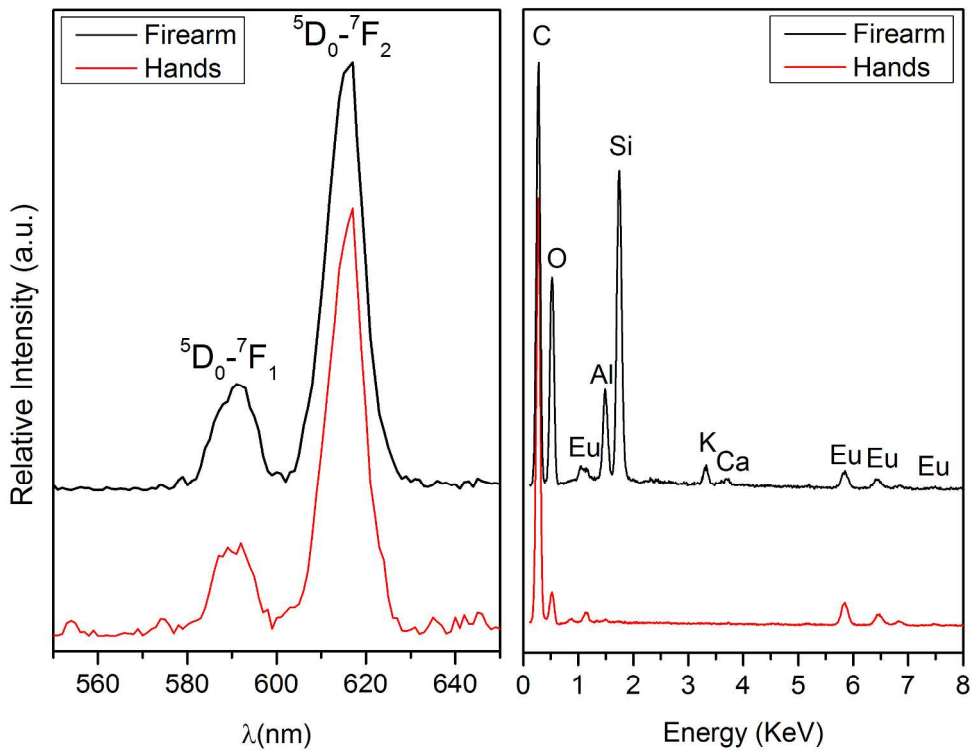


Figure 5. Emission spectra (left) and EDS (right) of the LGSR collected on the firearm (black) and shooter's hands (red) after shots with marked ammunition.

Figure 5
739x574mm (96 x 96 DPI)

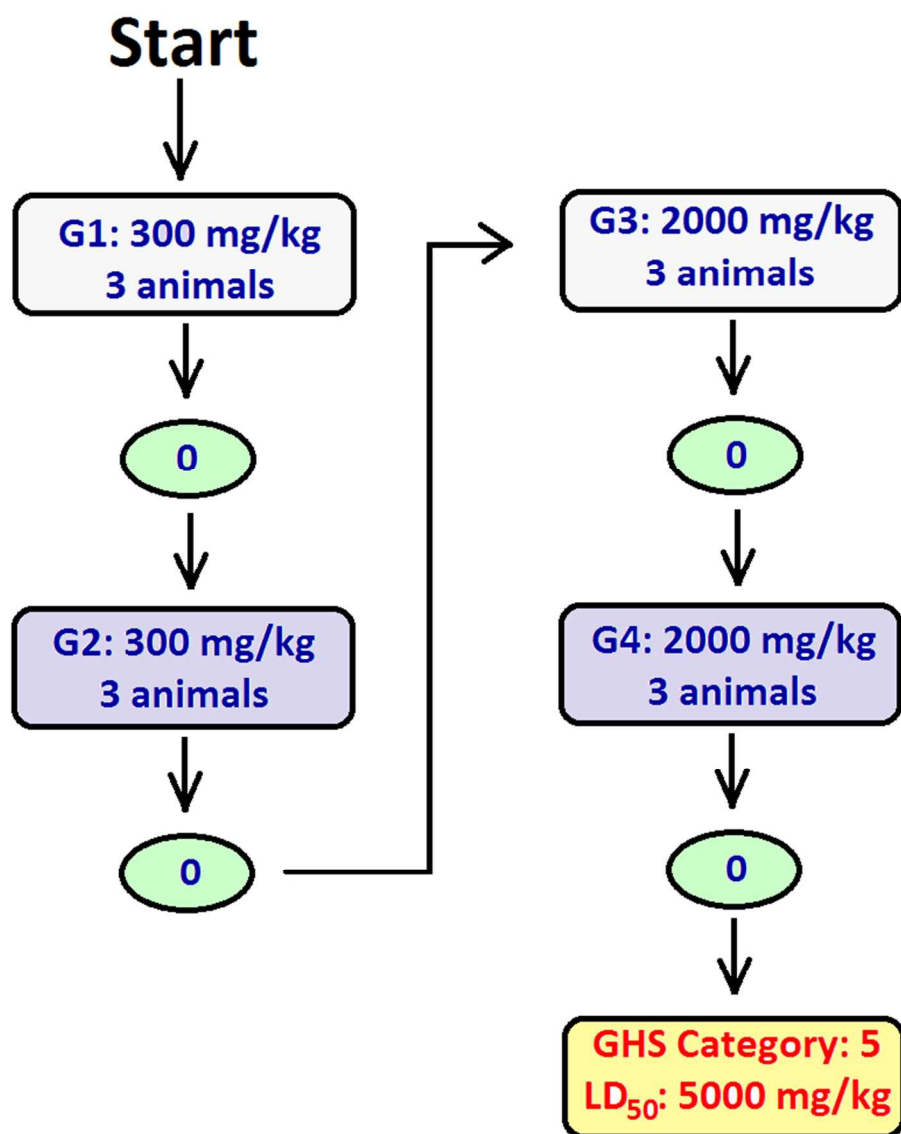


Figure 6. Steps followed by OECD guideline 423 to determine the LD₅₀ and GSH category. The circles indicate the number of dead or moribund animals by step.

Figure 6

223x272mm (96 x 96 DPI)

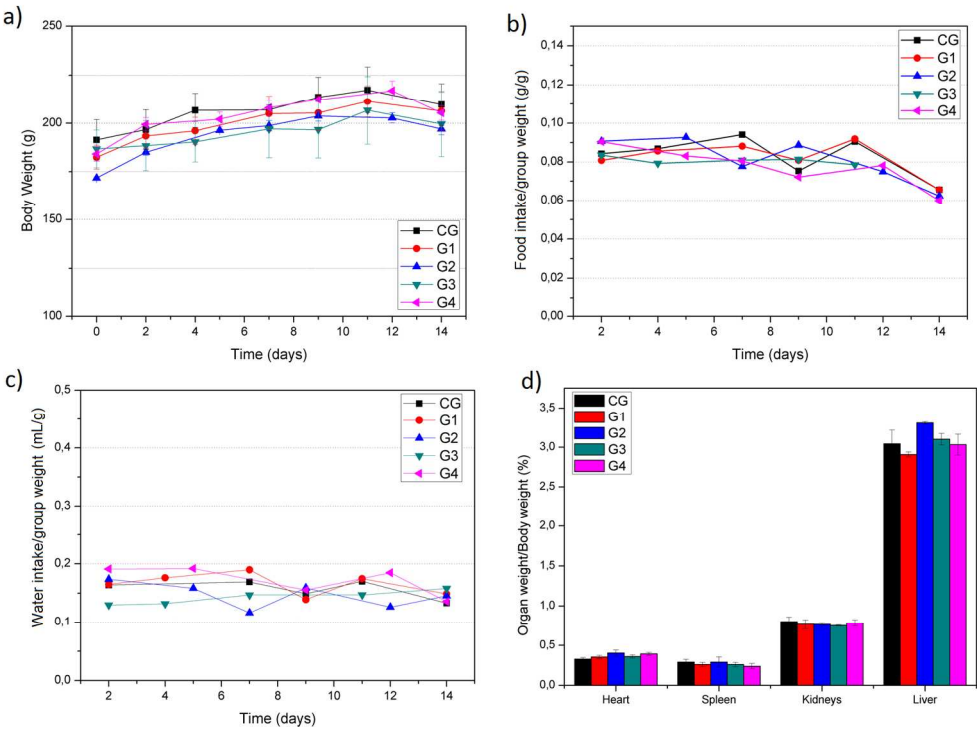


Figure 7. a) Evolution of the rat weight (mean $\pm$ SD), b) mean food and (c) water intake per group; d) organ weight as a function of the body weight in each group (mean $\pm$ SD).

Figure 7
488x357mm (96 x 96 DPI)

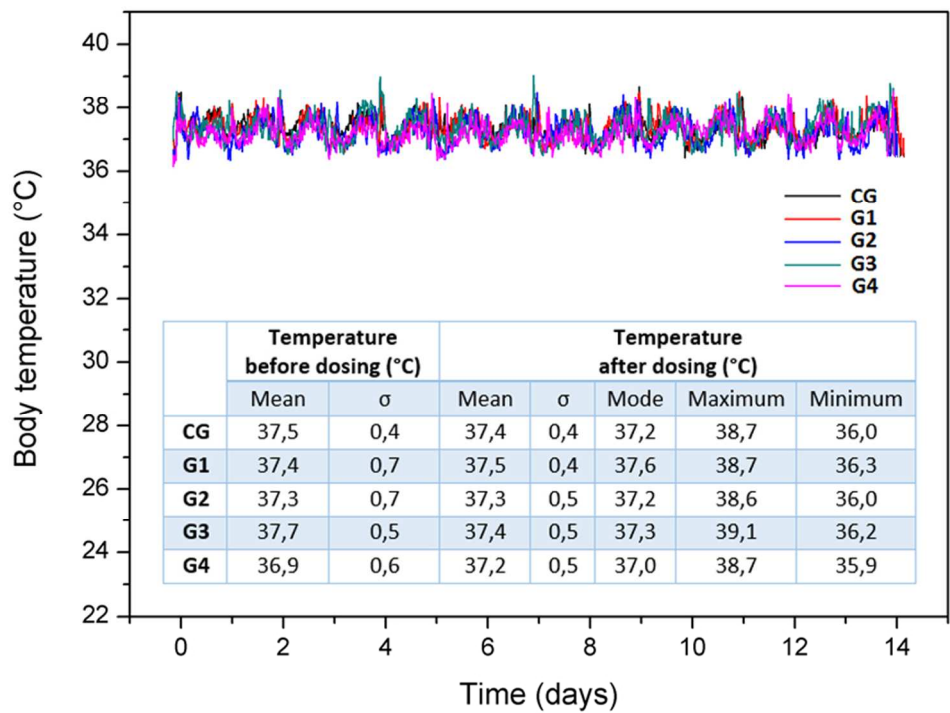


Figure 8. Time course of core body temperature for the G1, G2, G3, G4 and CG groups. The table shows the mean temperature of the groups 4 hours before dosing and 14 days after dosing; it also shows the maximum, minimum and the mode value temperatures after dosing.

Figure 8
241x178mm (96 x 96 DPI)

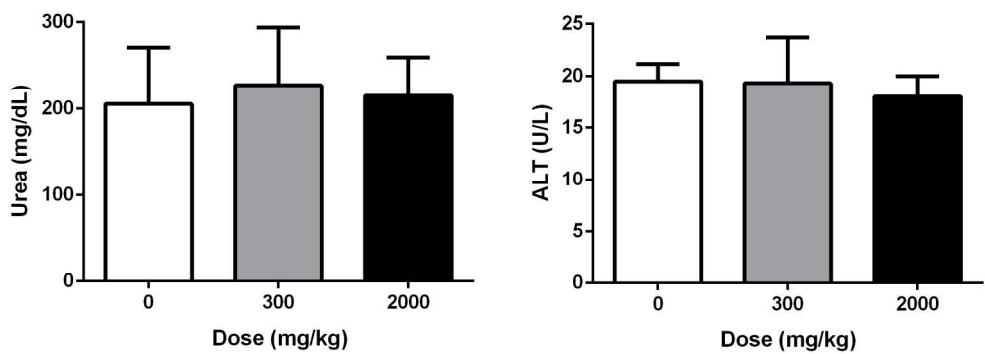


Figure 9. Biochemistry analysis of rats 14 days after treatment with the MOF [Eu(BTC)]. Values are expressed as the mean $\pm$ SEM. Dose = 0 mg/kg refers to the control group.

Figure 9
612x224mm (96 x 96 DPI)

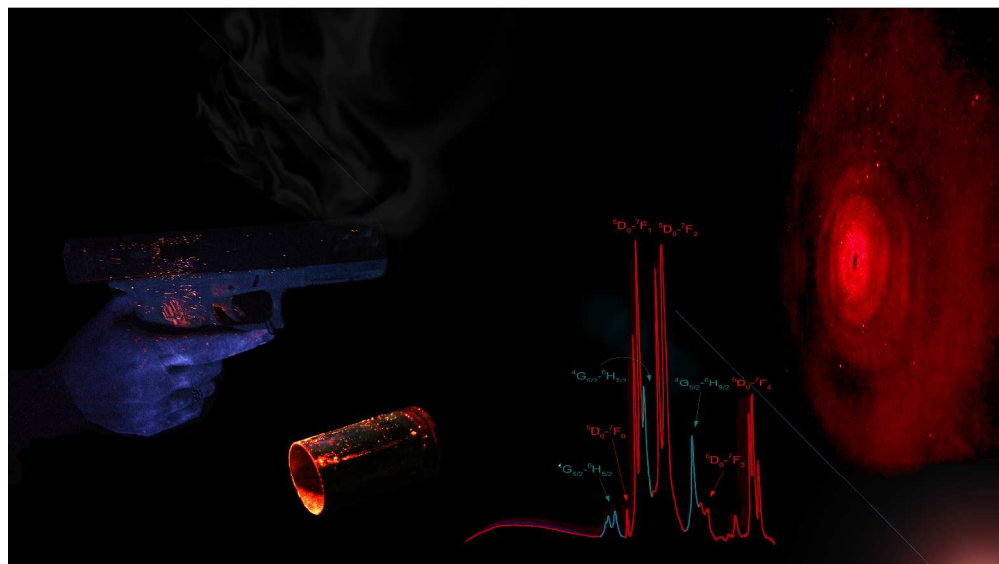


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1310x736mm (96 x 96 DPI)